COMPOSITIONS OF KOSZUL-REGULAR, H1-REGULAR, AND MIXED QUASI-REGULAR IMMERSIONS OF SPACES STAY IN CLASS

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 10655-10665 of sources/stacks-spaces-more-morphisms/pinned-original-source.tex; proof span is lines 10667-10670, the complete author proof (2 body lines).

Editorial note. Navigation stubs: 1 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Registration string 'composition-regular-immersion' appears in lane S12's algebra-lci-gysin cards as a prerequisite citation, not as a claim on this spaces label; gate occupancy confirmed FREE.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Divisors, lemma divisors-lemma-composition-regular-immersion (schemes side).

Source reference. Original lemma 067Q: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 10670–10670. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let S be a scheme. Let $i : Z \to Y$ and $j : Y \to X$ be immersions of algebraic spaces over S.*

(1) *If i and j are Koszul-regular immersions, so is $j \circ i$.*
(2) *If i and j are H_1-regular immersions, so is $j \circ i$.*
(3) *If i is an H_1-regular immersion and j is a quasi-regular immersion, then $j \circ i$ is a quasi-regular immersion.*

Proof. Immediate from the case of schemes, see Divisors, Lemma 067Q. □

Date: stacks-project:spaces-more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.